Referee Comment 3: Per-Segment vs Accumulated Sensitivity

Oct 30, 2026 · @Y Luo

The referee asks whether ING-Net's gain over the power-only baseline survives when many segments are combined. Our scaling test says it does not under linear stacking: the gain disappears within minutes at $\xi = 10^{-2}$ and within about a day at $\xi = 10^{-3}$. We recommend testing a nonlinear combination of per-segment posteriors, the approach the intermittent-background literature uses, before fixing the paper's positioning.

1. The referee's comment

The referee counts this among comments 1–8 that must be resolved before soundness can be assessed. The key passage:

> "The cross-correlation statistic accumulates as T_obs^(1/2), whereas the advantage demonstrated here is a per-segment effect governed by an estimator whose sampling variance the authors themselves quote as Var(kappa_4) ~ 24/N. Whether Delta survives stacking over N_seg >> 1 segments -- which is how every real SGWB search operates, and how Drasco-Flanagan, Thrane and Lawrence et al. pose the problem -- is never tested."

The referee asks for two things:

- **(a)** an explicit statement, in the abstract and at the head of Sec. V, that the limits are per-segment limits on a fixed dataset and not comparable to published broadband upper limits;
- **(b)** "at minimum a scaling analysis, and preferably a multi-segment result, showing that Delta does not shrink as segments are combined."

Comment 10 is linked: it asks for a non-Gaussian benchmark, naming the Drasco–Flanagan Gaussian-mixture likelihood and its implementation by Lawrence et al., both of which combine segments.

2. Why stacking erodes a per-segment kurtosis gain

The second-order and fourth-order channels improve at different rates as data accumulate, so any per-segment advantage of the kurtosis channel must eventually be overtaken. Hold fixed the segment length (4 s, n = 8192 samples), the duty cycle ξ and the noise level; let only the number of stacked segments N vary, and write σ_s^2 for the time-averaged signal variance per sample.

- **Second-order statistics** (cross-correlation, power): the signal shifts the statistic by about σ_s^2, while its noise falls as $1/\sqrt{nN}$. The detectable signal therefore scales as

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\sigma_s^2 \propto N^{-1/2}
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- **Excess kurtosis**: the signal shifts it by about $(c/\xi)\,\sigma_s^4$, and its noise also falls as $\sqrt{24/(nN)}$. Because the shift is quadratic in σ_s^2,

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\sigma_s^2 \propto \xi^{1/2}\, N^{-1/4}
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For a single segment the kurtosis channel wins through the $1/\xi$ factor. Each additional decade of N, however, moves the second-order limit down by 0.5 dex and the kurtosis limit by only 0.25 dex, so the gap closes by about 0.25 dex per decade. The crossover lies near a total sample count $nN \approx 27/\xi^2$. This is the referee's point made quantitative, and it holds for any estimator that combines segments by averaging their statistics.

3. Where the revised analysis stands per segment

Per segment, the gain is real and robust, but the absolute limits are far from where SGWB science is done. Values from the revised Table II (15 stretches, 5 per run from O3a, O3b and O4a, three training seeds each, transient-augmented training):

Duty cycle ξ	Gain Δ = Trad − ING-Net (dex)	Range over 15 stretches (dex)	Stretches with Δ ⩾ 0.1
10^{-3}	+0.50 ± 0.03	+0.30 to +0.85	15 of 15
10^{-2}	+0.23 ± 0.02	+0.06 to +0.35	14 of 15
⩾ 10^{-1}	−0.03 to +0.02	−0.14 to +0.12	none

- **Robustness checks already done:** the gain is unchanged by a 99% instead of 99.7% threshold, by coloured instead of white bursts (⩽ 0.03 dex), and by up to 47 times more network parameters (the gap does not shrink with capacity).
- **Absolute limits:** ING-Net reaches $\log_{10} \Omega_{\rm limit} = -4.7$ to -5.3 per 4 s segment at $\xi = 10^{-2}$ (after the calibration correction of comment 2). The LVK O3 isotropic search, integrating the whole run, gives $\Omega_{\rm GW} \leqslant 5.8 \times 10^{-9}$ for a flat spectrum, about 3.5 orders of magnitude deeper. The referee already found a two-order gap "conspicuous"; the corrected gap is larger.
- **What the manuscript says now:** only the Table II caption states that these are per-segment limits not comparable to broadband upper limits. The abstract, Sec. V and the discussion do not yet address stacking.

4. Scaling test with linear stacking (done 2026-10-26)

Under linear stacking the gain vanishes: within minutes at $\xi = 10^{-2}$, within about half a day at 10^{-3}, and within about a month at 10^{-4}.

Method. Synthetic, feature-level, no training. Unit-variance band-limited Gaussian noise plus the rev2 signal model (Poisson Hann-windowed bursts, $\tau = 8$ samples, 20–1000 Hz, H1–L1 antenna response with $\gamma(0) = -0.892$). Each feature ($C_{\rm HL}$, P, κ_H, κ_L) is averaged over N segments, as LVK averages per-segment estimates. Detection at 0.3% false-alarm probability uses the optimal linear (Fisher) combination of the stacked features for a known signal shape, the best any linear stacking of these features can do. Monte Carlo for $N \leqslant 30$, central-limit calculation for $N \geqslant 100$.

Self-checks (all passed). Second-order limits fall as N^(−1/2). At $\xi = 1$, $|\Delta| \leqslant 0.02$ for all N. At $N = 1$ the test reproduces ING-Net: $\Delta = 0.22$ versus 0.23 at $\xi = 10^{-2}$, and 0.58 versus 0.49 at 10^{-3}, so simple features with an optimal linear combination capture most of what ING-Net uses. Monte Carlo and analytic results agree to 0.03 at $N = 30$.

Gain Δ (dex) of cross-correlation + kurtosis over cross-correlation alone, against stacked duration:

ξ	4 s (N=1)	40 s	7 min	1.1 h	11 h	4.6 d	46 d	1.3 yr
10^{-2}	0.42	0.22	0.07	0.01	0	0	0	0
10^{-3}	0.85	0.66	0.45	0.25	0.10	0.01	0	0
3×10^{-4}	—	0.92	0.73	0.42	0.29	0.06	0.03	0.01
10^{-4}	—	1.28	0.92	0.65	0.48	0.21	0.10	0.01

ξ	4 s (N=1)	40 s	7 min	1.1 h	11 h	4.6 d	46 d	1.3 yr
10^{-1}	⩽ 0.08	0.01	0	0	0	0	0	0

—: at $\xi \leqslant 3 \times 10^{-4}$ most 4 s segments contain no event, so a single segment cannot reach 50% detection probability. With the power P added to both sides (noise level assumed known) the gains are a third to a half smaller (0.22 and 0.58 at N = 1 for $\xi = 10^{-2}$ and 10^{-3}; 0.92 at N = 10 for 10^{-4}) and vanish on the same time scales.

How firm this is. At large N every limit lies in the regime where individual events are far below the noise; there the non-Gaussian likelihood expanded to leading order is linear in the fourth cumulant, so linear stacking is close to optimal and "the gain vanishes for long stacks" is a firm conclusion. At small N and low ξ, where individual events are loud, a nonlinear combination (multiplying per-segment likelihoods, or the Drasco–Flanagan statistic) does better, so the entries there are lower bounds. The test uses ideal Gaussian noise and no frequency weighting of the cross-correlation.

5. How the literature exploits intermittency in long data

Long-duration integration is the standard setting for SGWB searches, and every paper the referee names works there; none averages statistics across segments as our test does. They multiply per-segment likelihoods instead.

Work	Setting	How segments are combined	Result relevant to us
Drasco & Flanagan 2003	Analytic; white noise, co-located detectors	Maximum-likelihood statistic of a two-component Gaussian mixture over all samples	Beats cross-correlation only when ξ is small enough; the threshold moves to smaller ξ as data accumulate. For ground-based data the advantage starts near $\xi \sim 10^{-3}$ and peaks at about a factor 2 in Ω near $\xi \sim 10^{-4}$. Consistent with our Section 4.

Work	Setting	How segments are combined	Result relevant to us
Smith & Thrane 2018	Binary-black-hole background, design sensitivity	Bayesian parameter estimation on all segments, with waveform templates	About one day of data to detect the BBH background, versus about 40 months with cross-correlation. The gain relies on templates that use each event's full signal, which our summaries do not.
Lawrence et al. 2023	Simulated data, 40,000 segments of 4 s (about 44 h), $\xi \approx 3 \times 10^{-3}$	Product over segments of a per-segment signal/noise mixture likelihood weighted by ξ	Outperforms the standard continuous cross-correlation search. Same segment length and duty-cycle range as ING-Net.

The lesson for us: our scaling test covers the least favourable way of combining segments. ING-Net already returns a calibrated per-segment posterior (comment 6), and with a uniform prior the product of per-segment posteriors is proportional to the product of likelihoods — the Lawrence et al. construction, with a learned likelihood in place of an analytic one. Whether that keeps a gain over hours to days has not been tested.

6. Tension with comment 2

The duty cycle at which ING-Net sees something no other search sees is the one where stacking erodes the gain fastest. The corrected single-event SNR (comment 2, revised Sec. V C) splits the prior range in two:

ξ	Events per 4 s segment	Single-event SNR at the limit	Visible to burst searches?	Gain lost under linear stacking after
10^{-3}	about 1	about 15 (12–22)	Yes: above cWB-type thresholds (network SNR ~ 8–10)	about half a day

ξ	Events per 4 s segment	Single-event SNR at the limit	Visible to burst searches?	Gain lost under linear stacking after
10^{-2}	about 10	about 3.9 (3.5–6.0)	No	a few minutes

- At $\xi = 10^{-3}$ the events at our limit are individually loud, so per segment we overlap with single-event burst searches; what we add there is a population-level posterior on (Ω, ξ) rather than a trigger list.
- At $\xi = 10^{-2}$ the events are individually invisible and only their combined kurtosis is detected. This was going to be the paper's distinctive regime, and it is exactly where accumulation removes the advantage within minutes.

The positioning therefore cannot simply be "ING-Net finds sub-threshold populations that other searches miss" without addressing accumulation.

7. Options

Three responses are possible; they differ in what the paper can claim and in how long the revision takes.

Option	What we do	What the paper can then claim	Cost	Main risk
(a) Report honestly	Add the referee's statement (a) to the abstract and Sec. V; add the scaling analysis of Section 4 as a new subsection; cite Drasco–Flanagan for comment 10 alongside our scaling table	A per-segment, short-time-scale detection and ξ-estimation method; long-duration gains left to nonlinear combination and lower ξ as future work	Writing only, 1–2 days	Satisfies the referee's minimum literally, but concedes the method is of limited use in the setting where SGWB searches actually operate

Option	What we do	What the paper can then claim	Cost	Main risk
(b) Test nonlinear combination	Combine per-segment ING-Net posteriors by multiplication (Section 8), compare with the power-only baseline combined the same way and with stacked cross-correlation	If the gain persists over hours to days at $\xi \leqslant 10^{-3}$: a per-segment non-Gaussian inference module that combines across segments, in line with the field. If not: option (a), now with evidence that nonlinear combination was tried	No training. About 1 day of code, a few hours on the server, 1–2 days of analysis and writing	Real evaluation noise covers only about 246 independent segments (16 min) per stretch; longer stacks need synthetic or recycled noise. Small per-segment calibration errors compound when N posteriors are multiplied
(c) Extend the prior	Retrain with $\log_{10} \xi$ down to −4 and evaluate there, targeting the regime where Drasco–Flanagan and our test find long-duration gains	A method aimed at very sparse backgrounds that keep an advantage after long integration	New work: retraining, evaluation and a scaling study, at least 1–2 weeks	Beyond a revision that patches the manuscript; at $\xi \sim 10^{-4}$ most segments are empty, so per-segment training becomes harder; lengthens the response

Options are not exclusive: (a) is needed in every case, (b) decides how (a) is framed, and (c) can be named as future work.

8. Recommendation and a draft design for option (b)

We recommend running (b) first and fixing the positioning on its outcome, while writing the statement and scaling analysis of (a) in any case. Either outcome gives a stronger answer than (a) alone: a positive result supports a long-duration role for the method, a negative one shows that the obvious remedy was tried. The design below would be fixed before any result is seen.

1. **Models and data.** The existing transient-augmented ING-Net and power-only networks of the three reference stretches (seeds 1400–1402); evaluation noise only, never seen in training. No retraining.
2. **Per-segment output.** For each 4 s segment, evaluate the log posterior density on a grid in ($\log_{10}\Omega$, $\log_{10}\xi$).
3. **Combination.** Sum the per-segment log posteriors over N segments. With the uniform prior this equals the summed log-likelihood up to a constant, for a stationary background with one (Ω, ξ) across segments.
4. **Detection and limits.** Use the combined posterior mean of $\log_{10}\Omega$ as the statistic; set the threshold at 0.3% false-alarm probability from noise-only stacks; find the 50%-detection amplitude by the same binary search as for single segments, injecting into every segment of the stack.
5. **Baselines.** The power-only network combined the same way, and linearly stacked cross-correlation.
6. **Stack lengths.** N = 1, 10, 30, 100 and about 250 from non-overlapping real evaluation noise (up to about 16 min per stretch). Longer stacks, to $N \sim 10^4$ (about 11 h), only with synthetic whitened Gaussian noise, after checking that single-segment calibration holds on it.
7. **Duty cycles.** $\xi = 10^{-3}$ and 10^{-2}.
8. **Calibration of the stack.** Check the coverage of the combined posterior by injection and recovery at each N, since small per-segment errors compound.
9. **Readout.** The combined gain $\Delta_comb(N)$ for each ξ, compared with the linear-stacking prediction of Section 4. A pre-stated criterion, for example "$\Delta_comb \geq 0.1$ dex at the longest stack", decides which framing the paper takes.

What (b) can and cannot show. With real noise alone it reaches about 16 minutes, where linear stacking already predicts a surviving gain of about 0.4 dex at $\xi = 10^{-3}$ but almost none at 10^{-2}. Hours or days are reachable only with synthetic noise, which weakens the result. Drasco–Flanagan bound the long-duration gain of any optimal statistic at about a factor 2–3 in Ω for weak events, so a large surviving gain at $\xi = 10^{-2}$ is not expected.

Time line if approved: design note fixed (half a day), code tested on synthetic data (1 day), server run (a few hours), analysis and manuscript text for (a) + (b) (1–2 days).

9. Decisions needed

1. **Run option (b) before fixing the positioning?** Recommended. It costs about four working days and no retraining.
2. **How far to push the stack length?** Real noise only (up to about 16 minutes, cleanest), or also synthetic noise to hours (longer reach, weaker evidence).
3. **Positioning if (b) is negative:** short-time-scale detection and ξ estimation, as in option (a); or should we consider option (c), a lower-ξ extension, as part of this revision rather than future work?
4. **Comment 10:** is citing Drasco–Flanagan alongside our scaling table enough, or do we implement the Drasco–Flanagan / Lawrence et al. mixture likelihood on our own data as a direct benchmark (extra work, but it would answer comment 10 fully and pairs naturally with option b)?

Everything else in the revision is written and does not depend on these choices; the abstract, the conclusion and Sec. VI C (relation to burst searches) wait for them.

Sources

- Referee report, DT18456 (received 2026-09-27), major comments 2, 3 and 10.
- Revised manuscript Table II and Sec. V C (rev2 analysis, 15 stretches); scaling test of 2026-10-26 (internal note and code).
- S. Drasco and É. É. Flanagan, Detecting a non-Gaussian stochastic background of gravitational radiation, Phys. Rev. D 67, 082003 (2003)
- R. Smith and E. Thrane, The optimal search for an astrophysical gravitational-wave background, Phys. Rev. X 8, 021019 (2018)
- J. Lawrence et al., A stochastic search for intermittent gravitational-wave backgrounds (2023)
- LIGO Scientific, Virgo and KAGRA Collaborations, Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo's third observing run, Phys. Rev. D 104, 022004 (2021)